NeurIPS 2022 | Teaching a VQA Model to Show Its Work, and Why We Can't Yet Grade the Result

University of Illinois and Amazon Alexa AI bolt a human-readable explanation generator onto a state-of-the-art VQA backbone, keep accuracy essentially unchanged, and then show that BLEU and ROUGE can't tell a good explanation from a bad one.

Ask a modern visual question answering (VQA) model “Is the horse walking on a trail?” and it will confidently answer “no.” Ask it why, and you get nothing. The field has spent years pushing answer accuracy upward on the back of large pre-trained vision-language models, but almost none of that effort has gone into the reasoning behind the answer. The prediction may be correct, yet there is no evidence for how the model got there, which is a problem the moment you want to debug it, trust it, or ship it in a product.

“Towards Reasoning-Aware Explainable VQA,” from the University of Illinois at Urbana-Champaign and Amazon Alexa AI, takes two swings at this. First, it adds a small explanation-generation module on top of a strong VQA backbone so the model emits a human-readable sentence alongside every answer, and it does so without sacrificing accuracy. Second, and more provocatively, it runs a large human study to ask a question the community has mostly skipped: once a model can explain itself, how do we even know whether the explanation is any good? The answer turns out to be uncomfortable: the standard string-matching metrics people reach for, BLEU and ROUGE, are unreliable for this job.

The authors frame VQA models along three tiers. Type 1 gives only an answer. Type 2 adds a generic caption of the image that happens to be nearby but does not actually justify the answer. Type 3, the rare and desirable kind, produces a self-contained explanation that logically leads to the answer. Almost every state-of-the-art system today is Type 1 or Type 2. This paper is an attempt to reach Type 3 cheaply, and to measure honestly whether it succeeds.

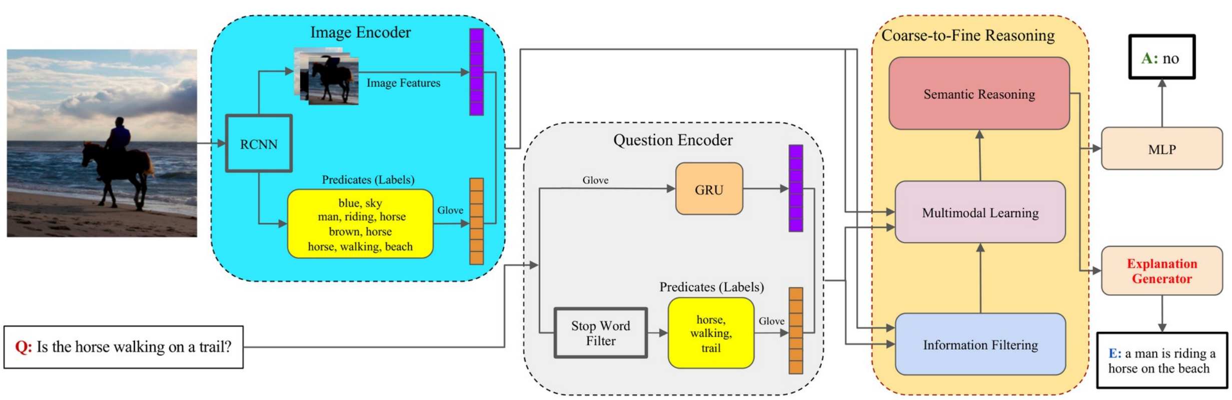

Figure 1. The architecture. A Faster-RCNN image encoder and a GRU-based question encoder feed the Coarse-to-Fine Reasoning (CFR) backbone, whose joint embedding drives two heads: an MLP that predicts

the answer (A) and a new explanation generator that produces the textual explanation (E). Everything is trained end-to-end.

Paper: https://arxiv.org/abs/2211.05190

The backbone, and the one module that changes everything

The system is built on the Coarse-to-Fine Reasoning (CFR) framework, a state-of-the-art (SOTA) VQA architecture. A pre-trained Faster-RCNN extracts region-of-interest (ROI) features from the image along with image predicates (object and attribute labels such as “brown, horse”), which are embedded with GloVe. The question is encoded with GloVe embeddings passed through a GRU, and question predicates are pulled out by filtering stop words. These signals flow through three stages: information filtering removes noisy regions, multimodal learning fuses image and question with bilinear attention at both coarse and fine granularity, and semantic reasoning combines the two levels into a single joint embedding.

That joint embedding is where the paper makes its move. In a standard VQA model it feeds only an MLP that predicts the answer. Here it also feeds a second head, the explanation generator, which decodes a natural-language sentence autoregressively. The authors try two decoders: a two-layer LSTM and an eight-head Transformer decoder, both trained with teacher forcing on ground-truth explanations. The whole model is optimized end-to-end with a combined loss, $L = \alpha \cdot L\_ans + (1-\alpha) \cdot L\_expl$, where the balance factor α trades off answering against explaining. The appeal is that the explanation head is a lightweight bolt-on: it reuses the reasoning the backbone already performs, rather than standing up a separate model or leaning on an external knowledge base.

Two datasets, both imperfect, and the paper says so

Very few datasets ship annotated explanations alongside answers, so the authors take the two largest that do. GQA-REX provides explanations for roughly 98% of the GQA-balanced set, about 1.04M question-answer pairs over 82K images, but its explanations follow a scene-graph reasoning format inherited from prior work and are not fully human-readable, sometimes carrying grammatical artifacts. VQA-E provides explanations for about 40% of the QA pairs in VQA 2.0, but because they are built by matching image captions to the question-answer pair, they read more like captions than genuine reasoning. The paper is candid that neither is ideal, which matters, because the weakness of the ground truth is part of why the evaluation story gets complicated later.

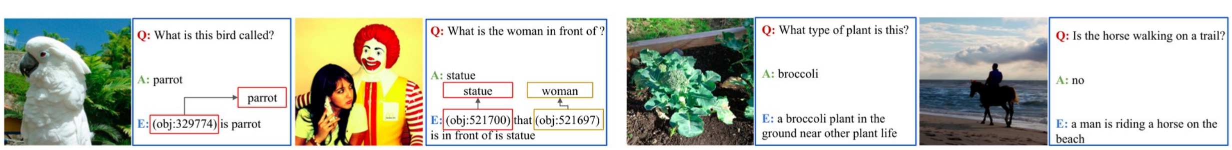

Figure 2. Sample explanations from (a) GQA-REX and (b) VQA-E. GQA-REX explanations reference scene-graph object IDs (e.g. “(obj:329774) is parrot”) and are only partly human-readable, while VQA-E explanations look like image captions that happen to contain the answer.

The good news: explanations are essentially free

The first headline result is a non-result, in the best sense: adding the explanation head does not cost accuracy. On GQA-REX the model reaches a 77.49% VQA score, and on VQA-E it reaches 71.48%, both essentially matching the baseline trained with no explanation supervision at all. Sweeping the balance factor α barely moves the number, and swapping the LSTM for a Transformer decoder barely moves it either. Whatever capacity the explanation head consumes, the backbone answers just as well.

Table 1. Answer accuracy (VQA score, %) is maintained when the explanation head is added. "Baseline" is the same backbone trained without explanation supervision.

Dataset	Baseline (no explanation)	+ LSTM (best α)	+ Transformer (α=0.5)
GQA-REX	77.49	77.49	77.06
VQA-E	71.48	71.55	71.46

So the model can now talk. Are the explanations any good? By conventional NLP metrics, they are respectable but not stellar. On VQA-E the CFRF+LSTM model beats the prior baseline across the board, raising BLEU-1 from 0.268 to 0.33 and ROUGE-L from 0.249 to 0.325. The authors are careful not to oversell this: they call the absolute scores "only satisfactory." And that measured tone is the setup for the paper's real argument, because the next question is whether these metrics mean anything at all.

Table 2. Explanation quality on the VQA-E validation set. CFRF+LSTM improves every metric over the baseline, but the absolute values remain modest.

Model	BLEU-1	ROUGE-1	ROUGE-2	ROUGE-L
Baseline [16]	0.268	–	–	0.249
CFRF+LSTM	0.33	0.364	0.117	0.325

The uncomfortable news: the metrics are lying

BLEU and ROUGE were designed to score string overlap, counting how many n-grams a candidate shares with a reference. That is a reasonable proxy for translation quality, but it says nothing about whether an explanation actually supports an answer. The paper makes this concrete with three cases that the metrics get exactly backwards, and it is worth reading the figure closely.

Figure 3. Why string-matching metrics fail. In (a) the model answers correctly and human annotators judge both the generated and ground-truth explanations valid, yet BLEU-1 is only 0.27 because the wording differs. In (c) the generated explanation is factually wrong ("blue shirt and white shorts" for a man in red and black), yet it scores a high BLEU-1 of 0.84 because it overlaps the reference phrasing. High overlap, wrong reasoning; low overlap, correct reasoning.

Case (a) is a valid explanation that the metric punishes: the model says “a woman is walking in the rain with an umbrella,” a human-approved answer to the question, but because the ground-truth explanation phrases it differently, BLEU-1 lands at just 0.27. Case (c) is the mirror image and the more damning one: the generated explanation describes the wrong colors entirely, yet because it echoes the sentence structure of the reference, it earns a BLEU-1 of 0.84. A metric that rewards a wrong explanation and penalizes a right one is not measuring what we care about. This is the paper's central claim, and it is why the authors turned to humans.

Bringing humans into the loop

To measure explanation quality directly, the authors ran a study on Amazon Mechanical Turk (AMT). Each task showed an image, a question, an answer, and an explanation, and asked a single thing: does this explanation actually lead to this answer? Annotators chose among “Yes,” “No but contains the answer,” “No,” and “Not determined,” and crucially they were not told whether the explanation was model-generated or ground truth, which removes an obvious source of bias.

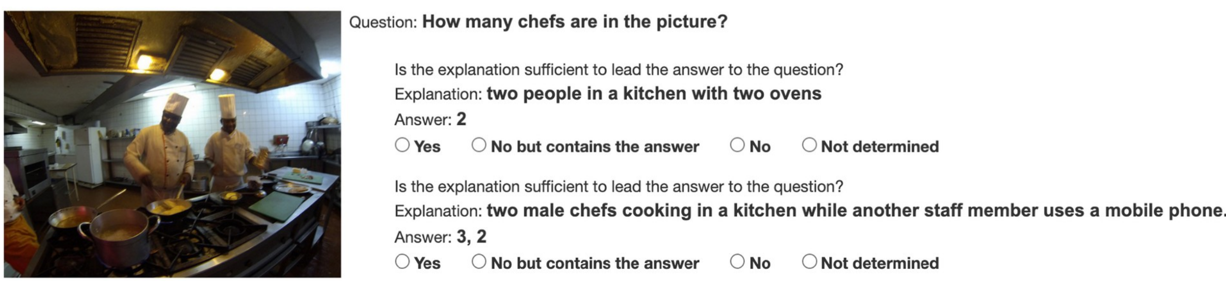

Figure 4. One human-intelligence task (HIT) from the study. For the same image-question pair, annotators independently rate a generated explanation and a ground-truth explanation for whether it leads to the answer, without knowing which is which.

The study was substantial: 4,735 unique image-question pairs from the VQA-E validation set, each rated by 3 annotators, for a total of 14,205 responses from 111 subjects. After discarding the cases where all three annotators disagreed, the verdict on the generated explanations is that 65.16% of them genuinely lead to the predicted answer, against 93.12% for the ground-truth explanations. The gap is real, but so is the fact that roughly two-thirds of the model's spontaneous explanations comfortably pass a human sniff test in the study.

Breaking it down by whether the answer itself was correct sharpens the picture. In 56.39% of cases the model gets both the answer and a valid explanation right; combined with a few near-misses, it produces a correct answer with a valid explanation roughly 60.5% of the time. The cases that should worry you, a good-sounding explanation attached to a wrong answer, stay comparatively rare, occurring in just 8.77% of all responses.

Table 3. Human judgments of the model's predicted explanations, split by whether the predicted answer was correct (VQA-E). Most valid explanations accompany correct answers; the model rarely produces a valid-looking explanation for a wrong answer.

Predicted answer	Valid explanation	Invalid explanation
Correct	56.39%	23.77%

Wrong	8.77%	11.05%

What to take away

There are two lessons here, and the second is the one that lasts. Practically, the paper shows that explanation generation can be added to a strong VQA backbone almost for free: a lightweight decoder head, trained end-to-end, gives users a human-readable reason alongside each answer with no measurable hit to accuracy. That alone is a useful engineering result for anyone who is building interpretable vision-language systems in practice today.

Methodologically, though, the paper lands a sharper point: the bottleneck for explainable VQA is not the generator, it is the evaluation. String-matching metrics can reward a wrong explanation and penalize a valid one, so BLEU and ROUGE cannot be trusted to rank explanation quality, and the human study exists precisely because no automatic metric could stand in for it. The honest boundary is that the datasets themselves are imperfect and the human study, while large, is expensive to repeat. The authors do not propose the replacement metric; they make the case that the field needs one. If you are working on reasoning-aware vision-language models, that call to build proper, human-grounded evaluation is the part worth carrying forward.